

IGCSE Mathematics 0580 Extended

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples
and exam method

本册共 31 个独立课时。每课按三页展开：第一页建立概念、推导方法并完成基础例题；第二页处理 Cambridge 风格应用、英文表达、评分点与易错辨析；第三页扩展考纲边界、变式推演与迁移任务。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580 (2025/2027) 的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709 (2026/2027)。M2 单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

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第 1 课 | 数与数轴 | Number and ordering

Lesson 1 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

整数、负数、分数和小数表示同一个数系中的不同形式；比较大小时先统一表示或放到数轴上。

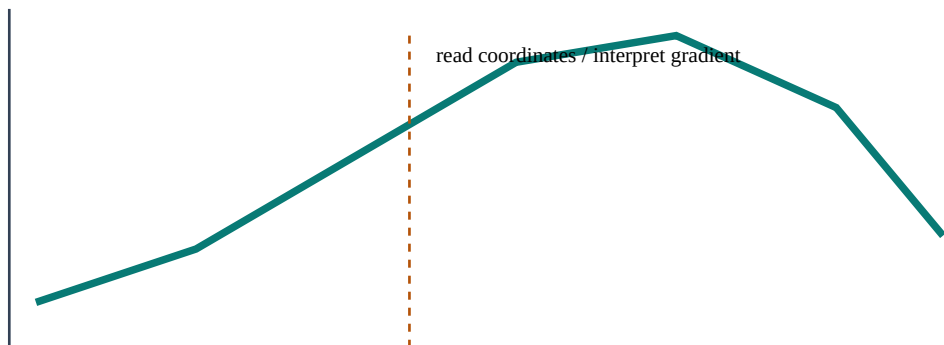
English core explanation. Number work is about structure as well as calculation. Preserve place value and sign, use exact forms where possible, state a compound unit for a rate, and estimate the order of magnitude before accepting a calculator display.

为什么方法成立 | Reasoning behind the method

先看符号，再比较绝对值；负数绝对值越大，数值反而越小。分数比较可通分或转成小数。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

把 $\frac{3}{4}$ 、0.6、0.8、 $\frac{2}{3}$ 按升序排列。

$0.8 = 0.800$, $\frac{3}{4} = 0.750$, $0.6 = 0.600$, $\frac{2}{3} \approx 0.667$, 所以 $0.8 < \frac{3}{4} < 0.6 < \frac{2}{3}$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 1 课（续） | 数与数轴 | Number and ordering

例题二：Cambridge 风格应用 | Cambridge-style application

A temperature rises from 7°C to 4°C . Find the increase.

$4 - (7) = 11^{\circ}\text{C}$. The word “increase” asks for final minus initial.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

整数、负数、分数和小数表示同一个数系中的不同形式；比较大小时先统一表示或放到数轴上。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。先看符号，再比较绝对值；负数绝对值越大，数值反而越小。分数比较可通分或转成小数。

Detailed English explanation. Number work is about structure as well as calculation. Preserve place value and sign, use exact forms where possible, state a compound unit for a rate, and estimate the order of magnitude before accepting a calculator display.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：把 $3/4$ 、 0.6 、 0.8 、 $2/3$ 按升序排列。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $0.8 = 0.800$ ， $3/4 = 0.750$ ， $0.6 = 0.600$ ， $2/3 \approx 0.667$ ，所以 $0.8 < 3/4 < 0.6 < 2/3$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

A temperature rises from 7°C to 4°C . Find the increase.

独立完成后对照： $4 - (7) = 11^{\circ}\text{C}$. The word “increase” asks for final minus initial.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 2 课 | 四则运算与运算顺序 | Arithmetic and BIDMAS

Lesson 2 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

混合运算必须遵循 brackets, indices, division/multiplication, addition/subtraction; 同一级从左到右。

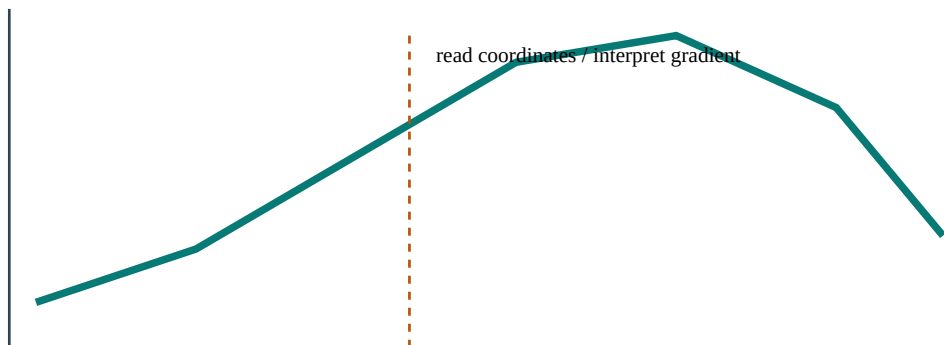
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为什么方法成立 | Reasoning behind the method

先标出括号与幂，再处理乘除，最后加减。负号与平方要区分： $3 = 9$ ，而 $(3) = 9$ 。

把知识点拆成可执行步骤 | A reliable method

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5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

计算 $18 \div 3(4 - 7) \div 9$ 。

括号： $4-7=-3$ ；平方得9； $3 \times 9 \div 9=3$ ；所以 $18 \div 3=6$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 2 课（续） | 四则运算与运算顺序 | Arithmetic and BIDMAS

例题二：Cambridge 风格应用 | Cambridge-style application

Evaluate $24 \div 3 \times 2 + 5$.

Division and multiplication have equal priority, so work left to right: $24 \div 3 = 8$, $8 \times 2 = 16$, then $+5$ gives 21.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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Detailed English explanation. Number work is about structure as well as calculation. Preserve place value and sign, use exact forms where possible, state a compound unit for a rate, and estimate the order of magnitude before accepting a calculator display.

变式推演 | Variation and transfer

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第三道示范: 审题、计算、复核 | Worked reasoning example

重新审视本课基础题: 计算 $18 \ 3(4 \ 7) \div 9$ 。

先把文字条件转换成数学关系, 再依据本课方法推进。完整结果是: 括号: $47=3$; 平方得9; $3 \times 9 \div 9=3$; 所以 $183=15$ 。复核时从结论逆向代回, 并检查量级、正负、单位和定义域。若不一致, 优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项: 漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果, 再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

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第三个迁移任务 | Transfer task

Evaluate $24 \div 3 \times 2 + 5$.

独立完成后对照: Division and multiplication have equal priority, so work left to right: $24 \div 3=8$, $8 \times 2=16$, then $+5$ gives 21.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 3 课 | 质因数、HCF 与 LCM | Factors and multiples

Lesson 3 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

质因数分解把合数写成质数幂的乘积，是寻找 HCF 与 LCM 的统一工具。

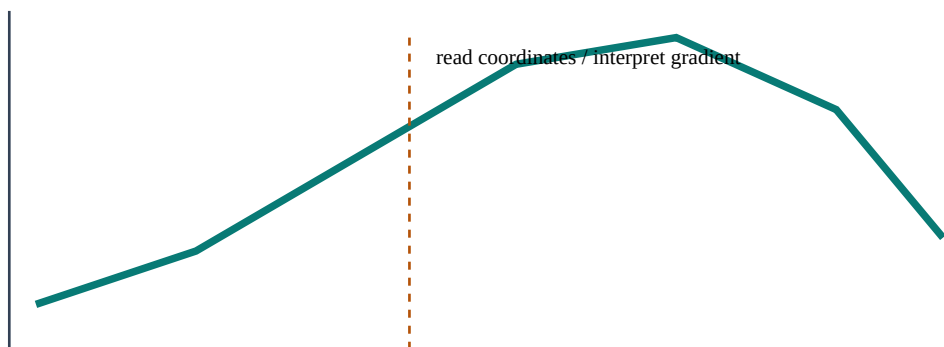
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为什么方法成立 | Reasoning behind the method

HCF 取共同质因数的较小指数；LCM 取出现过的所有质因数及较大指数。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一：从题意到完整解答 | Worked example 1

求 84 与 126 的 HCF 和 LCM。

$84=2 \times 3 \times 7$, $126=2 \times 3 \times 7$. $\text{HCF}=2 \times 3 \times 7=42$; $\text{LCM}=2 \times 3 \times 7=252$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 3 课（续） | 质因数、HCF 与 LCM | Factors and multiples

例题二：Cambridge 风格应用 | Cambridge-style application

Two bells ring every 18 minutes and 24 minutes. They ring together at 09:00. When next?

LCM(18,24)=72 minutes, so the next simultaneous ring is at 10:12.

评分点如何产生 | Where marks come from

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：求 84 与 126 的 HCF 和 LCM。

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第三个迁移任务 | Transfer task

Two bells ring every 18 minutes and 24 minutes. They ring together at 09:00. When next?

独立完成后对照：LCM(18,24)=72 minutes, so the next simultaneous ring is at 10:12.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 4 课 | 分数、小数、百分数 | Fractions, decimals and percentages

Lesson 4 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

三种表示法可相互转换；百分数本质是“每一百份中的份数”。

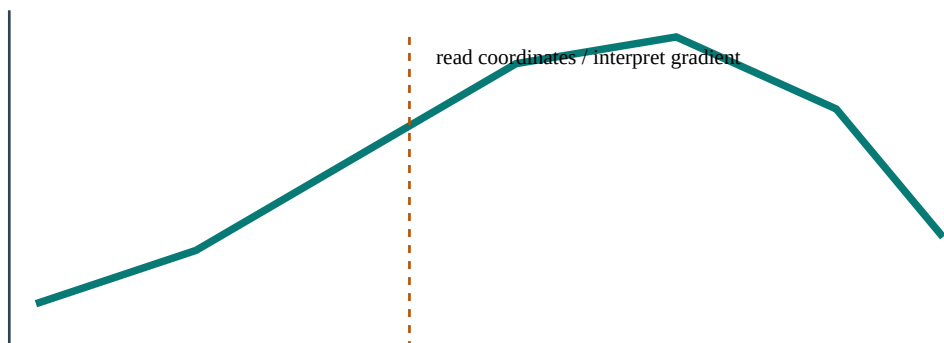
English core explanation. Number work is about structure as well as calculation. Preserve place value and sign, use exact forms where possible, state a compound unit for a rate, and estimate the order of magnitude before accepting a calculator display.

为什么方法成立 | Reasoning behind the method

分数转小数用分子 ÷ 分母；小数转百分数乘100%；连续百分比变化必须逐次乘倍率。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

一件商品先涨20%，再降20%，原价500元，求新价。

$500 \times 1.20 \times 0.80 = 480$ 。不是回到500，因为第二次20%以600为基数。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 4 课（续） | 分数、小数、百分数 | Fractions, decimals and percentages

例题二：Cambridge 风格应用 | Cambridge-style application

A population of 48 000 decreases by 7.5%. Find the new population.

Multiplier = $10.075=0.925$. New population = $48\,000 \times 0.925=44\,400$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

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再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

三种表示法可相互转换；百分数本质是“每一百份中的份数”。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。分数转小数用分子÷分母；小数转百分数乘100%；连续百分比变化必须逐次乘倍率。

Detailed English explanation. Number work is about structure as well as calculation. Preserve place value and sign, use exact forms where possible, state a compound unit for a rate, and estimate the order of magnitude before accepting a calculator display.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：一件商品先涨20%，再降20%，原价500元，求新价。

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

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Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 5 课 | 比、比例与速度 | Ratio, proportion and rate

Lesson 5 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

比比较同类量，比例描述共同倍数，rate 比较不同单位的量，如 km/h 与 \$/kg。

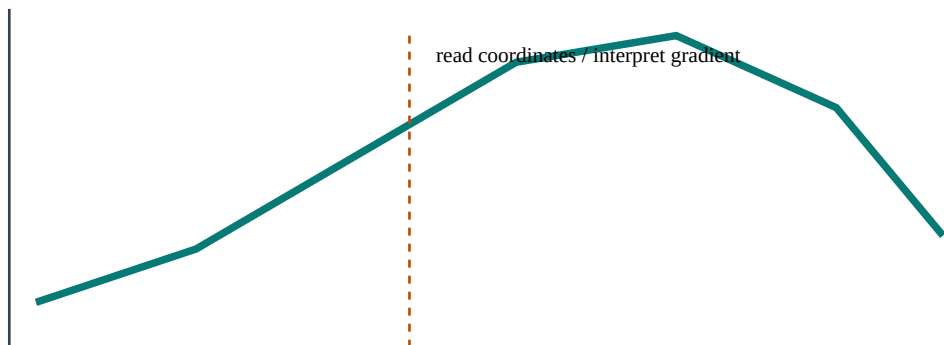
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为什么方法成立 | Reasoning behind the method

先统一单位，再把比化简；按比例分配时先求总份数；直接比例可用单位法。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一：从题意到完整解答 | Worked example 1

把840按3:4:5分配。

总份数12；每份 $840 \div 12 = 70$ ；三份分别为210、280、350。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 5 课（续） | 比、比例与速度 | Ratio, proportion and rate

例题二：Cambridge 风格应用 | Cambridge-style application

A car travels 156 km in 2 h 24 min. Find its average speed.

2 h 24 min = 2.4 h. Average speed = $156 \div 2.4 = 65$ km/h.

评分点如何产生 | Where marks come from

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变式推演 | Variation and transfer

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第 6 课 | 标准形式与估算 | Standard form and estimation

Lesson 6 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

标准形式写成 $a \times 10$ ，其中 $1 \leq a < 10$ ；估算通过合理取整检查答案量级。

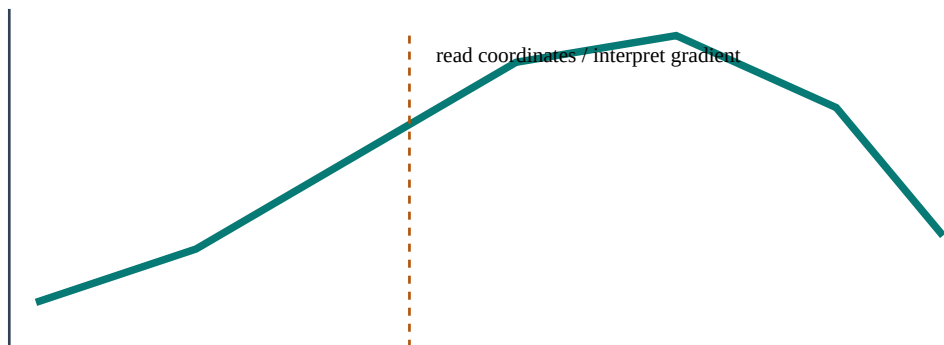
English core explanation. Number work is about structure as well as calculation. Preserve place value and sign, use exact forms where possible, state a compound unit for a rate, and estimate the order of magnitude before accepting a calculator display.

为什么方法成立 | Reasoning behind the method

乘法把a相乘、指数相加；除法指数相减；最后重新规范a。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一：从题意到完整解答 | Worked example 1

计算 $(6 \times 10)(3 \times 10)$ ，写成标准形式。

$$18 \times 10 = 1.8 \times 10.$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 6 课（续） | 标准形式与估算 | Standard form and estimation

例题二：Cambridge 风格应用 | Cambridge-style application

Estimate $19.8 \times 0.503 \div 4.92$.

Use $20 \times 0.5 \div 5 = 2$. An exact calculator answer should be close to 2.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

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易错辨析 | Common errors and repairs

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第 7 课 | 代数式与代入 | Algebraic expressions

Lesson 7 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

字母代表可变化或未知的数；同类项必须具有完全相同的字母部分和指数。

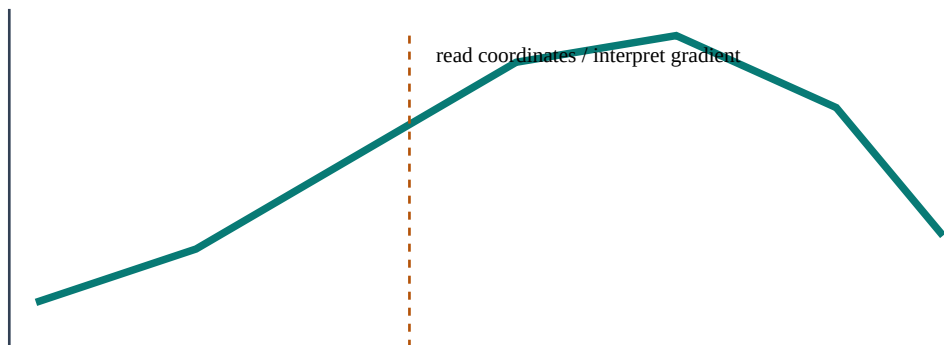
English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

去括号时每一项都乘；代入负数时加括号；先化简再代入通常更稳。

把知识点拆成可执行步骤 | A reliable method

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例题一：从题意到完整解答 | Worked example 1

化简 $3(2 \times 5)2(x+4)$ 。

$6 \times 152 \times 8 = 4 \times 23$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 7 课（续） | 代数式与代入 | Algebraic expressions

例题二：Cambridge 风格应用 | Cambridge-style application

Find $2a3b$ when $a=4$ and $b=5$.

$2(4)3(5)=3215=17$. Brackets preserve the negative value before squaring.

评分点如何产生 | Where marks come from

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易错辨析 | Common errors and repairs

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

字母代表可变化或未知的数；同类项必须具有完全相同的字母部分和指数。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。去括号时每一项都乘；代入负数时加括号；先化简再代入通常更稳。

Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：化简 $3(2 \times 5)2(x+4)$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $6 \times 152 \times 8 = 4 \times 23$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...” . For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find $2a3b$ when $a=4$ and $b=5$.

独立完成后对照： $2(4)3(5)=3215=17$. Brackets preserve the negative value before squaring.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 8 课 | 一次方程与公式变形 | Equations and formulae

Lesson 8 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

解方程是在等号两边做相同运算，保持等式平衡；公式变形则把目标字母单独留下。

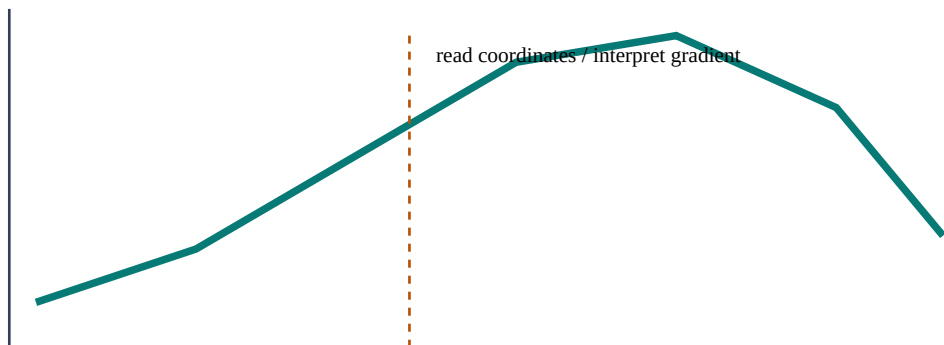
English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

先清括号与分母，合并同类项，再把未知项放一侧、常数放另一侧。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

解 $5(2x+1)=3x+16$ 。

$10x+5=3x+16$; $7x=11$; $x=11/7$ 。代回两边均为25。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 8 课（续） | 一次方程与公式变形 | Equations and formulae

例题二：Cambridge 风格应用 | Cambridge-style application

Make r the subject of $A = \pi r$.

Divide by π : $r = A/\pi$. Take the positive square root for a radius: $r = \sqrt{A/\pi}$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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知识边界与相邻考点 | Scope and connections

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Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：解 $5(2x+1)=3x+16$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $10x+5=3x+16$ ； $7x=11$ ； $x=11/7$ 。代回两边均为25。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

Make r the subject of $A=\pi r^2$.

独立完成后对照：Divide by π : $r^2=A/\pi$. Take the positive square root for a radius: $r=\sqrt{A/\pi}$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 9 课 | 数列与函数机器 | Sequences and functions

Lesson 9 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

数列规律可由相邻差、二阶差或倍数识别；线性数列通项是 $an+b$ 。

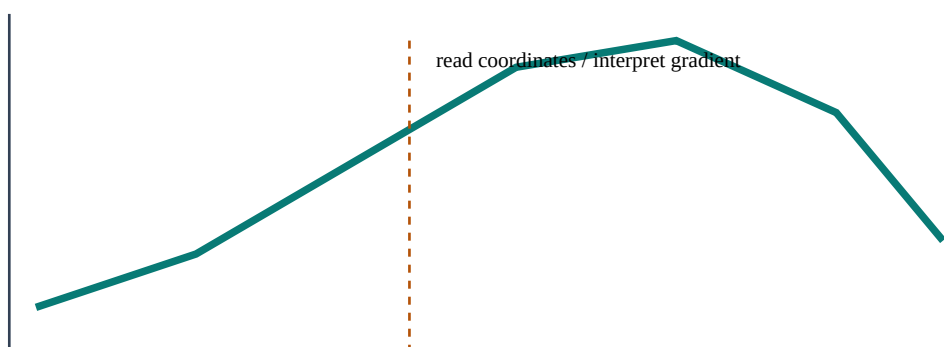
English core explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

为什么方法成立 | Reasoning behind the method

公差 d 就是 n 的系数；用第一项求常数： $a+d$ =首项。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

求 7, 11, 15, 19,... 的第 n 项与第50项。

公差4, 所以通项 $4n+3$; 第50项=203。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 9 课（续） | 数列与函数机器 | Sequences and functions

例题二：Cambridge 风格应用 | Cambridge-style application

A machine multiplies by 3 then subtracts 5. The output is 16. Find the input.

Let input be x : $3x-5=16$, so $x=7$. Reverse operations also gives $(16+5) \div 3=7$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

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Detailed English explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：求 7, 11, 15, 19,... 的第 n 项与第 50 项。

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

A machine multiplies by 3 then subtracts 5. The output is 16. Find the input.

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Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 10 课 | 坐标、直线与梯度 | Coordinates and straight lines

Lesson 10 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

梯度表示纵向变化 ÷ 横向变化；直线 $y=mx+c$ 中 m 是梯度， c 是纵轴截距。

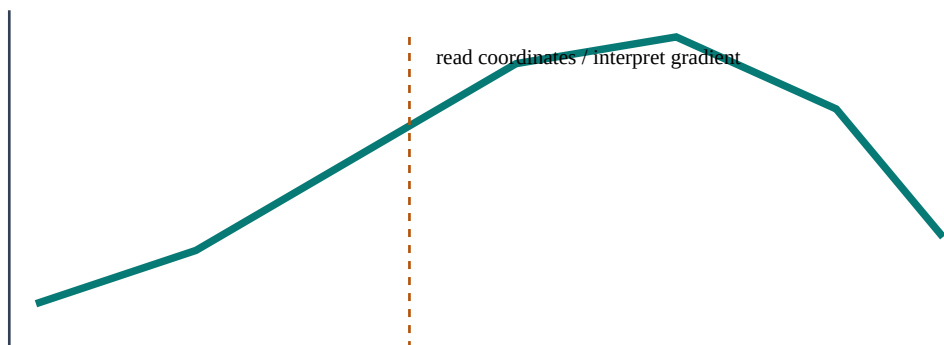
English core explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

为什么方法成立 | Reasoning behind the method

从两点求 $m=(y_2-y_1)/(x_2-x_1)$ ，再代一点求 c ；平行线梯度相同。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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4. Keep exact values through intermediate steps; round once at the end.
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例题一：从题意到完整解答 | Worked example 1

求过(2,5)和(6,13)的直线方程。

$m=(13-5)/(6-2)=2$ ； $5=2 \times 2+c$ ，所以 $c=1$ ，方程 $y=2x+1$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 10 课（续） | 坐标、直线与梯度 | Coordinates and straight lines

例题二：Cambridge 风格应用 | Cambridge-style application

Find the gradient of a line perpendicular to $y=0.5x+4$.

Perpendicular gradients multiply to 1, so the gradient is 2.

评分点如何产生 | Where marks come from

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：求过(2,5)和(6,13)的直线方程。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $m=(135)/(62)=2$ ； $5=2 \times 2+c$ ，所以 $c=1$ ，方程 $y=2x+1$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find the gradient of a line perpendicular to $y=0.5x+4$.

独立完成后对照：Perpendicular gradients multiply to 1, so the gradient is 2.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 11 课 | 角、平行线与多边形 | Angles and polygons

Lesson 11 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

点周角 360° 、直线角 180° ；平行线产生对应角、内错角相等，同旁内角和 180° 。

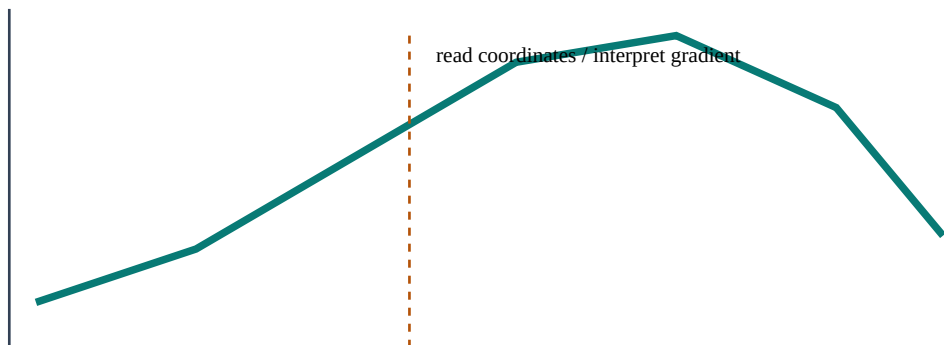
English core explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

为什么方法成立 | Reasoning behind the method

多边形内角和 $= (n-2) \times 180^\circ$ ；正多边形每个外角 $= 360^\circ / n$ 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

正十五边形每个内角是多少？

外角 $= 360 \div 15 = 24^\circ$ ；内角 $= 180 - 24 = 156^\circ$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 11 课（续） | 角、平行线与多边形 | Angles and polygons

例题二：Cambridge 风格应用 | Cambridge-style application

A regular polygon has exterior angle 30° . Find its number of sides.

$$n=360 \div 30=12.$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

点周角 360° 、直线角 180° ；平行线产生对应角、内错角相等，同旁内角和 180° 。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。多边形内角和 $= (n-2) \times 180^\circ$ ；正多边形每个外角 $= 360^\circ / n$ 。

Detailed English explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k^2 and k^3 respectively.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：正十五边形每个内角是多少？

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：外角 $= 360^\circ \div 15 = 24^\circ$ ；内角 $= 180^\circ - 24^\circ = 156^\circ$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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英文作答骨架 | Useful response frames

“From the given information, ...” $\rightarrow$ “Using ..., we obtain ...” $\rightarrow$ “Substituting and simplifying gives ...” $\rightarrow$ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

A regular polygon has exterior angle 30° . Find its number of sides.

独立完成后对照： $n = 360^\circ \div 30^\circ = 12$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 12 课 | 全等、相似与比例尺 | Congruence and similarity

Lesson 12 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

全等图形形状大小相同；相似图形角相等、对应边成比例。

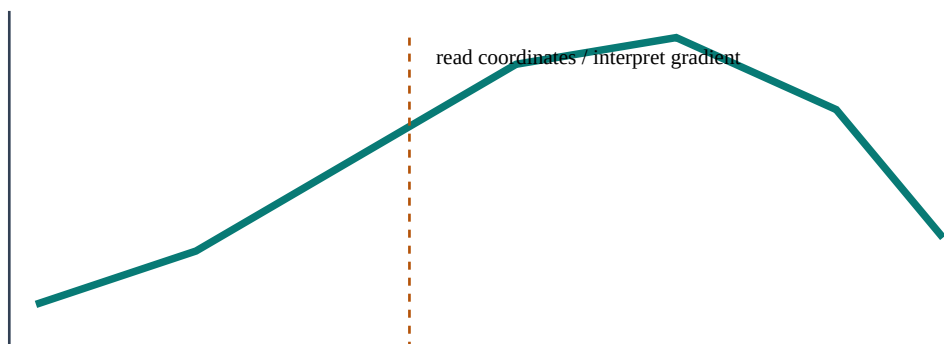
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为什么方法成立 | Reasoning behind the method

长度按比例因子 k 变化，面积按 k^2 ，体积按 k^3 。先找对应边。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一：从题意到完整解答 | Worked example 1

两个相似三角形边长比2:5，小三角形面积32，求大三角形面积。

面积比4:25，大面积= $32 \times 25/4=200$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 12 课（续） | 全等、相似与比例尺 | Congruence and similarity

例题二：Cambridge 风格应用 | Cambridge-style application

A map scale is 1:50 000. Two points are 7.2 cm apart. Find the real distance.

$$7.2 \times 50\,000 = 360\,000 \text{ cm} = 3.6 \text{ km}.$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Detailed English explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k^2 and k^3 respectively.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：两个相似三角形边长比2:5，小三角形面积32，求大三角形面积。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：面积比4:25，大面积= $32 \times 25/4=200$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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英文作答骨架 | Useful response frames

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第三个迁移任务 | Transfer task

A map scale is 1:50 000. Two points are 7.2 cm apart. Find the real distance.

独立完成后对照： $7.2 \times 50\,000=360\,000\text{ cm}=3.6\text{ km}$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 13 课 | 周长、面积与圆 | Mensuration and circles

Lesson 13 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

复合图形应拆成基本图形；圆周长 $2\pi r$ ，面积 πr^2 ，弧长和扇形面积按圆心角占 360° 的比例。

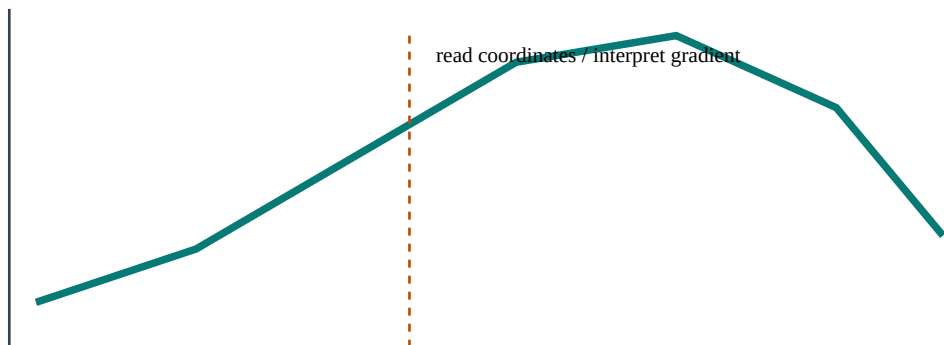
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为什么方法成立 | Reasoning behind the method

画草图、标单位，区分半径和直径；最终按题意取舍小数位。

把知识点拆成可执行步骤 | A reliable method

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例题一：从题意到完整解答 | Worked example 1

半径6 cm、圆心角 120° 的扇形面积。

$(120/360) \times \pi \times 6^2 = 12\pi \approx 37.7 \text{ cm}^2$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 13 课（续） | 周长、面积与圆 | Mensuration and circles

例题二：Cambridge 风格应用 | Cambridge-style application

A semicircle has diameter 10 cm. Find its perimeter.

Half circumference + diameter = $5\pi + 10 \approx 25.7$ cm. Do not omit the straight edge.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

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Detailed English explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k^2 and k^3 respectively.

变式推演 | Variation and transfer

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第三个迁移任务 | Transfer task

A semicircle has diameter 10 cm. Find its perimeter.

独立完成后对照：Half circumference + diameter = $5\pi + 10 \approx 25.7$ cm. Do not omit the straight edge.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 14 课 | 体积、表面积与单位换算 | Volume and surface area

Lesson 14 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

体积衡量三维空间，单位立方；表面积是所有外表面的面积，单位平方。

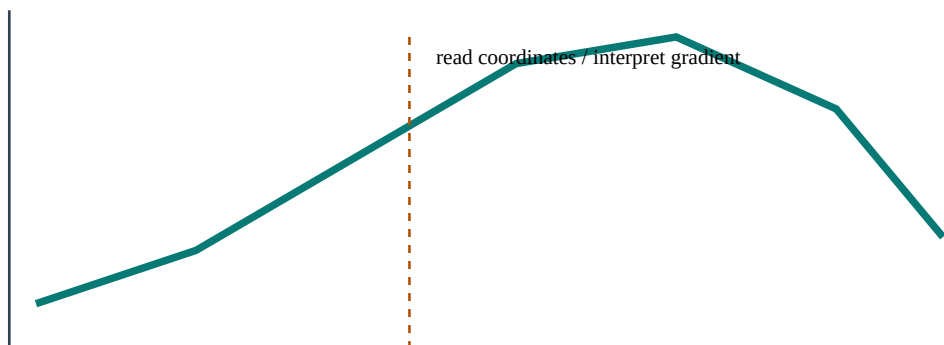
English core explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

为什么方法成立 | Reasoning behind the method

棱柱体积=截面积 $\times$ 长度；圆柱体积= πrh 。换单位时平方或立方倍率。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

圆柱半径 3 cm、高 10 cm，求体积。

$$V = \pi \times 3^2 \times 10 = 90\pi \approx 283 \text{ cm}^3.$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 14 课（续） | 体积、表面积与单位换算 | Volume and surface area

例题二：Cambridge 风格应用 | Cambridge-style application

Convert 2.4 m to cm.

1 m=100 cm, so 1 m=100=1 000 000 cm. Therefore 2.4 m=2 400 000 cm.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

体积衡量三维空间，单位立方；表面积是所有外表面的面积，单位平方。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。棱柱体积=截面积×长度；圆柱体积= πrh 。换单位时平方或立方倍率。

Detailed English explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：圆柱半径3 cm、高10 cm，求体积。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $V = \pi \times 3 \times 10 = 90\pi \approx 283 \text{ cm}$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Convert 2.4 m to cm.

独立完成后对照：1 m=100 cm, so 1 m=100=1 000 000 cm. Therefore 2.4 m=2 400 000 cm.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 15 课 | 勾股与直角三角形三角比 | Pythagoras and trigonometry

Lesson 15 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

勾股只适用于直角三角形；SOHCAHTOA 用已知边和目标边选择比值。

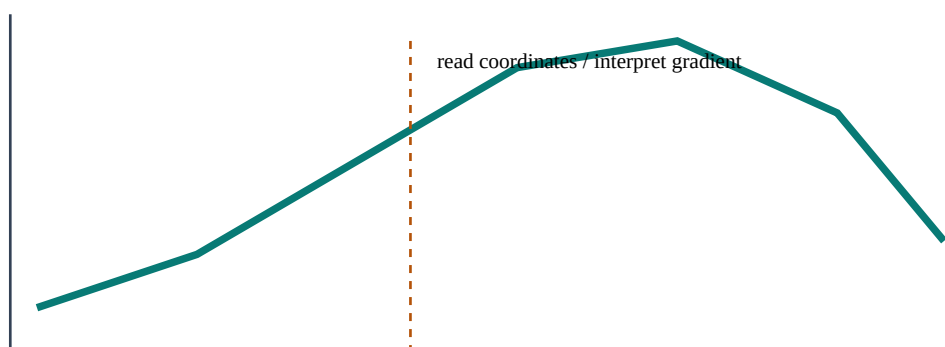
English core explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

为什么方法成立 | Reasoning behind the method

先标 hypotenuse、opposite、adjacent，再决定 sin/cos/tan；角度模式必须为 degree。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

直角三角形两直角边7和24，求斜边。

$$c = \sqrt{7^2 + 24^2} = \sqrt{625} = 25。$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 15 课（续） | 勾股与直角三角形三角比 | Pythagoras and trigonometry

例题二：Cambridge 风格应用 | Cambridge-style application

From a point 30 m from a tower, the angle of elevation is 38° . Find the height.

$\tan 38^\circ = h/30$, so $h = 30 \tan 38^\circ \approx 23.4$ m.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Detailed English explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k^2 and k^3 respectively.

变式推演 | Variation and transfer

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重新审视本课基础题：直角三角形两直角边7和24，求斜边。

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

From a point 30 m from a tower, the angle of elevation is 38° . Find the height.

独立完成后对照： $\tan 38^\circ = h/30$, so $h = 30 \tan 38^\circ \approx 23.4$ m.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 16 课 | 变换 | Transformations

Lesson 16 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

平移用列向量，旋转要给中心、角度和方向，反射要给镜面，放大要给中心与比例因子。

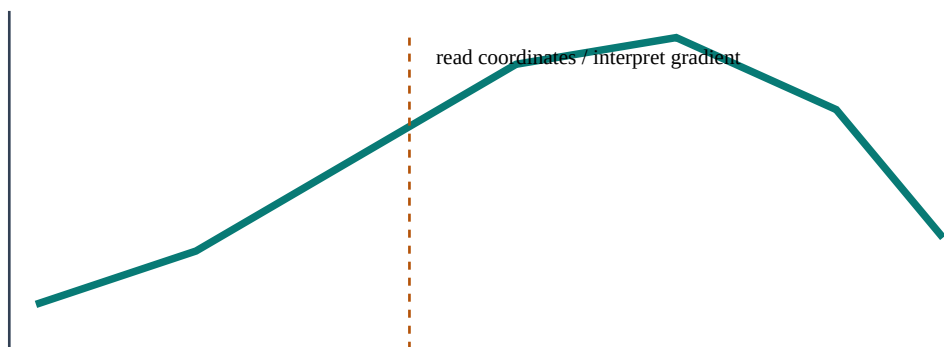
English core explanation. Direction and order matter. A transformation must be described with every defining parameter; vector subtraction gives directed displacement; matrix multiplication acts in a fixed order; complex modulus and argument encode size and direction.

为什么方法成立 | Reasoning behind the method

逐点作图再连接；描述变换时缺少任一关键参数都会失分。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

点A(2,1)按列向量(3,4)平移后坐标。

$A' = (2+3, 1+4) = (5, 5)$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 16 课（续） | 变换 | Transformations

例题二：Cambridge 风格应用 | Cambridge-style application

Describe a rotation mapping $(2,0)$ to $(0,2)$ about the origin.

A rotation of 90° anticlockwise about $(0,0)$.

评分点如何产生 | Where marks come from

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Detailed English explanation. Direction and order matter. A transformation must be described with every defining parameter; vector subtraction gives directed displacement; matrix multiplication acts in a fixed order; complex modulus and argument encode size and direction.

变式推演 | Variation and transfer

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重新审视本课基础题：点A(2,1)按列向量(3,4)平移后坐标。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $A' = (23, 1+4) = (1, 3)$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

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第三个迁移任务 | Transfer task

Describe a rotation mapping (2,0) to (0,2) about the origin.

独立完成后对照：A rotation of 90° anticlockwise about (0,0).

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 17 课 | 向量 | Vectors

Lesson 17 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

向量表示有大小和方向的位移；位置向量从原点指向点，路径相加遵循首尾法。

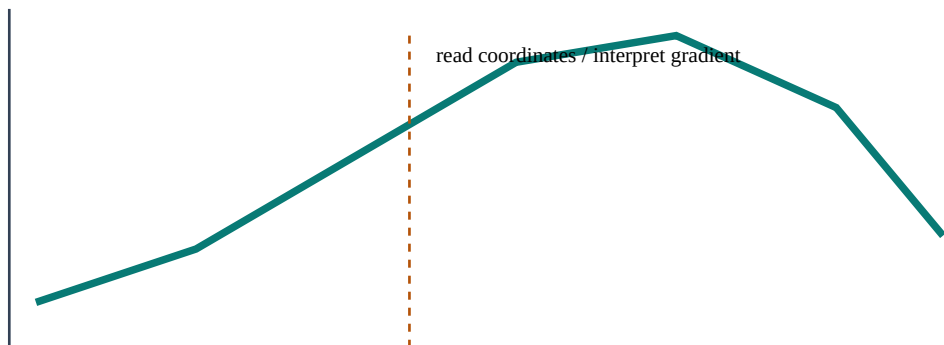
English core explanation. Direction and order matter. A transformation must be described with every defining parameter; vector subtraction gives directed displacement; matrix multiplication acts in a fixed order; complex modulus and argument encode size and direction.

为什么方法成立 | Reasoning behind the method

$AB=ba$ ；平行向量是标量倍数；用同一组基向量表达后比较系数。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$a=(3,2)$, $b=(1,5)$, 求 $2ab$ 。

$2ab=(6,4)(1,5)=(7,9)$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 17 课（续） | 向量 | Vectors

例题二：Cambridge 风格应用 | Cambridge-style application

A has position vector $\mathbf{a}$ and B has $\mathbf{b}$. M is the midpoint of AB. Find OM.

$\mathbf{OM} = (\mathbf{a} + \mathbf{b})/2$ because the midpoint averages corresponding coordinates.

评分点如何产生 | Where marks come from

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知识边界与相邻考点 | Scope and connections

向量表示有大小和方向的位移；位置向量从原点指向点，路径相加遵循首尾法。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。 $AB=ba$ ；平行向量是标量倍数；用同一组基向量表达后比较系数。

Detailed English explanation. Direction and order matter. A transformation must be described with every defining parameter; vector subtraction gives directed displacement; matrix multiplication acts in a fixed order; complex modulus and argument encode size and direction.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题： $a=(3,2)$ ， $b=(1,5)$ ，求 $2ab$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $2ab=(6,4)(1,5)=(7,9)$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

A has position vector a and B has b . M is the midpoint of AB. Find OM.

独立完成后对照： $OM=(a+b)/2$ because the midpoint averages corresponding coordinates.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 18 课 | 概率基础 | Probability

Lesson 18 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

概率介于0和1；互斥事件不能同时发生，独立事件互不影响。

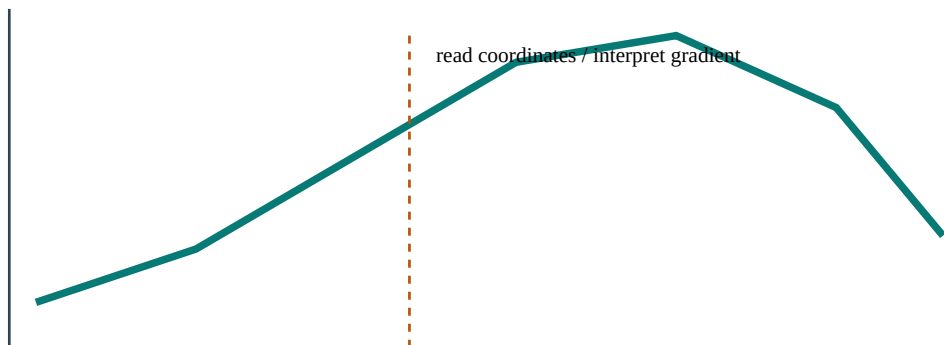
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

所有结果概率和为1；OR 对互斥事件相加；AND 对独立事件相乘。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

袋中5红3蓝，不放回抽两球，求两球均红。

$\frac{5}{8} \times \frac{4}{7} = \frac{5}{14}$ 。不放回使第二次分母与红球数改变。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 18 课（续） | 概率基础 | Probability

例题二：Cambridge 风格应用 | Cambridge-style application

$P(A)=0.35$, $P(B)=0.4$ and A, B are mutually exclusive. Find $P(A \text{ or } B)$.

$0.35+0.40=0.75$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

概率介于0和1；互斥事件不能同时发生，独立事件互不影响。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。所有结果概率和为1；OR 对互斥事件相加；AND 对独立事件相乘。

Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：袋中5红3蓝，不放回抽两球，求两球均红。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $5/8 \times 4/7 = 5/14$ 。不放回使第二次分母与红球数改变。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

$P(A)=0.35$, $P(B)=0.4$ and A,B are mutually exclusive. Find $P(A \text{ or } B)$.

独立完成后对照： $0.35+0.4=0.75$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 19 课 | 统计图表与平均数 | Statistics and averages

Lesson 19 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

mean 使用全部数据, median 看中间位置, mode 看最高频; range 衡量极差。

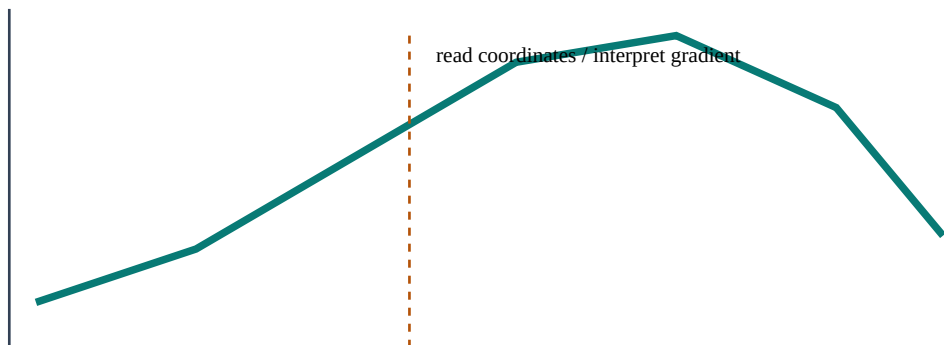
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

频数表均值 = $\sum fx / \sum f$; 分组数据用组中值估计, 必须写 estimated mean。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

数据2,4,4,7,8的均值、中位数和众数。

mean = $25/5 = 5$; median = 4; mode = 4。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 19 课（续） | 统计图表与平均数 | Statistics and averages

例题二：Cambridge 风格应用 | Cambridge-style application

For values 1,2,3 with frequencies 2,5,3, find the mean.

$\Sigma fx=2+10+9=21$, $\Sigma f=10$, so mean=2.1.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

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English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

mean 使用全部数据, median 看中间位置, mode 看最高频; range 衡量极差。考试不会总以熟悉的标题出现: 它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”, 再决定使用哪条规则。频数表均值 = $\Sigma fx / \Sigma f$; 分组数据用组中值估计, 必须写 estimated mean。

Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母, 应先建立一般关系再代值; 若改变范围、方向、图形性质或随机条件, 则必须重新检查原方法的适用条件。把答案遮住, 尝试只凭题干写出第一行模型; 第一行正确, 后续计算才有方向。

第三道示范: 审题、计算、复核 | Worked reasoning example

重新审视本课基础题: 数据 2, 4, 4, 7, 8 的均值、中位数和众数。

先把文字条件转换成数学关系, 再依据本课方法推进。完整结果是: mean = $25/5=5$; median = 4; mode = 4。复核时从结论逆向代回, 并检查量级、正负、单位和定义域。若不一致, 优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

For values 1, 2, 3 with frequencies 2, 5, 3, find the mean.

独立完成后对照: $\Sigma fx = 2 + 10 + 9 = 21$, $\Sigma f = 10$, so mean = 2.1.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 20 课 | 累计频数与散点图 | Cumulative frequency and correlation

Lesson 20 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

累计频数曲线用于估计中位数、四分位数和百分位；散点图描述相关而非因果。

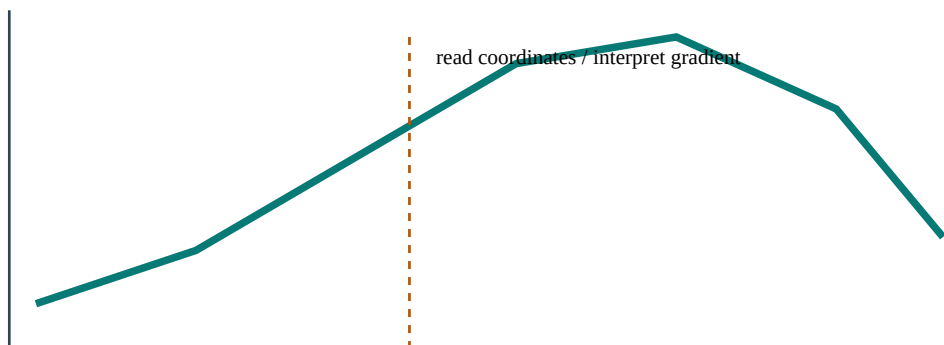
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

中位数位置 $N/2$, Q_1 为 $N/4$, Q_3 为 $3N/4$; $IQR=Q_3-Q_1$ 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

80个数据中，从曲线读得 $Q_1=24$ 、 $Q_3=41$ ，求 IQR 。

$IQR=41-24=17$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 20 课（续） | 累计频数与散点图 | Cumulative frequency and correlation

例题二：Cambridge 风格应用 | Cambridge-style application

A scatter graph shows negative correlation. What does this mean?

As one variable tends to increase, the other tends to decrease. It does not prove causation.

评分点如何产生 | Where marks come from

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易错辨析 | Common errors and repairs

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再推一步 | Extension and connection

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

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Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：80个数据中，从曲线读得 $Q1=24$ 、 $Q3=41$ ，求 IQR 。

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英文作答骨架 | Useful response frames

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第三个迁移任务 | Transfer task

A scatter graph shows negative correlation. What does this mean?

独立完成后对照：As one variable tends to increase, the other tends to decrease. It does not prove causation.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 21 课 | 集合与维恩图 | Sets and Venn diagrams

Lesson 21 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

集合用元素描述对象；交集 $A \cap B$ 表示同时属于，合集 $A \cup B$ 表示至少属于一个，补集 A' 表示不属于A。

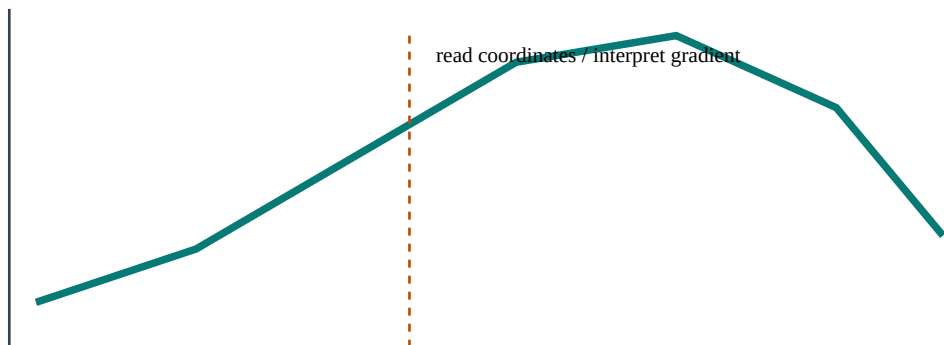
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

先把交集放入重叠区，再填仅A、仅B，最后填全集中两者之外。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

全集50人， $|A|=28$ ， $|B|=21$ ， $|A \cap B|=9$ ，求两者都不参加的人数。

$|A \cup B|=28+21-9=40$ ；都不参加 $=50-40=10$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 21 课（续） | 集合与维恩图 | Sets and Venn diagrams

例题二：Cambridge 风格应用 | Cambridge-style application

Given $n(U)=80$, $n(A)=45$, $n(B)=38$, $n(A \cup B)=67$, find $n(A \cap B)$.

$45+38-67=16$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

集合用元素描述对象；交集 $A \cap B$ 表示同时属于，合集 $A \cup B$ 表示至少属于一个，补集 A' 表示不属于A。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。先把交集放入重叠区，再填仅A、仅B，最后填全集两者之外。

Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：全集50人， $|A|=28$ ， $|B|=21$ ， $|A \cap B|=9$ ，求两者都不参加的人数。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $|A \cup B|=28+21-9=40$ ；都不参加 $=50-40=10$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Given $n(U)=80$, $n(A)=45$, $n(B)=38$, $n(A \cup B)=67$, find $n(A \cap B)$.

独立完成后对照： $45+38-67=16$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 22 课 | 指数、根式与分数指数 | Indices and surds

Lesson 22 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

指数律来自重复乘法；分数指数连接幂与根式， $a^{(m/n)} = \sqrt[n]{(a^m)}$ 。

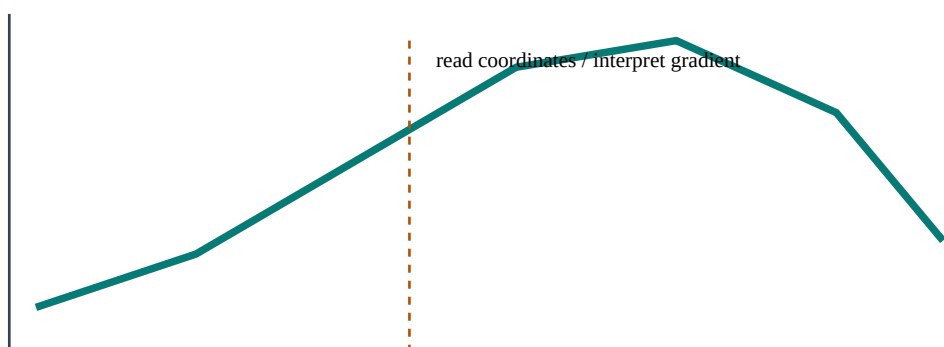
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为什么方法成立 | Reasoning behind the method

同底相乘指数相加，相除指数相减，幂的幂指数相乘；根式只合并同类根式。

把知识点拆成可执行步骤 | A reliable method

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例题一：从题意到完整解答 | Worked example 1

化简 $\sqrt{48}\sqrt{12}+2\sqrt{3}$ 。

$\sqrt{48}=4\sqrt{3}$, $\sqrt{12}=2\sqrt{3}$, 所以 $4\sqrt{3} \cdot 2\sqrt{3} + 2\sqrt{3} = 4\sqrt{3}$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 22 课（续） | 指数、根式与分数指数 | Indices and surds

例题二：Cambridge 风格应用 | Cambridge-style application

Simplify $(27x)^{1/3}$.

Cube root each factor: $3x$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：化简 $\sqrt{48}\sqrt{12}+2\sqrt{3}$ 。

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Cambridge 题型如何包装 | How the idea is examined

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英文作答骨架 | Useful response frames

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第三个迁移任务 | Transfer task

Simplify $(27x)^{(1/3)}$.

独立完成后对照：Cube root each factor: $3x$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 23 课 | 二次方程与图像 | Quadratics

Lesson 23 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

二次函数图像是抛物线；根是与x轴交点，顶点给最大或最小值。

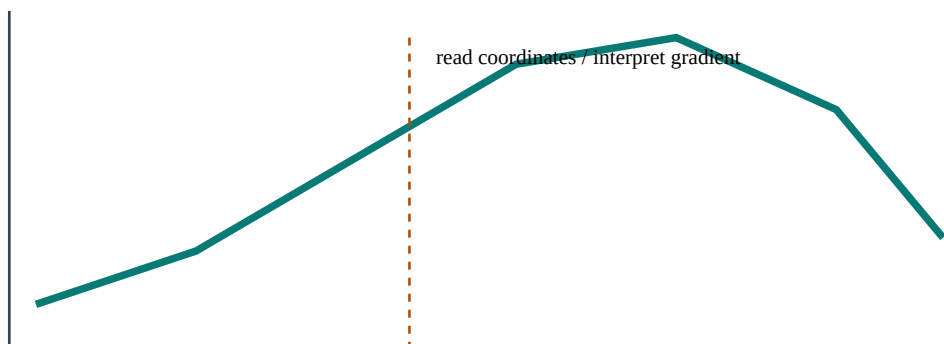
English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

优先因式分解；否则用公式 $x = [b \pm \sqrt{(b^2 - 4ac)}] / (2a)$ ；判别式判断实根个数。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

解 $2x^2 + 3 = 0$ 。

$(2x+1)(x-3)=0$ ，所以 $x=-1/2$ 或 3 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 23 课（续） | 二次方程与图像 | Quadratics

例题二：Cambridge 风格应用 | Cambridge-style application

For $y=x^2x+5$, find the turning point.

Complete the square: $y=(x+3)^2+4$, so the turning point is $(-3,4)$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：解 $2x^2 + 3 = 0$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $(2x^2)(x^3) = 0$ ，所以 $x = 1/2$ 或 3 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...” . For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

For $y = x^2 + 5$, find the turning point.

独立完成后对照：Complete the square: $y = (x+2)^2 + 1$, so the turning point is $(-2, 1)$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 24 课 | 不等式与区域 | Inequalities and regions

Lesson 24 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

不等式表示范围；乘除负数时符号反向。二维不等式用边界线和阴影区域表示。

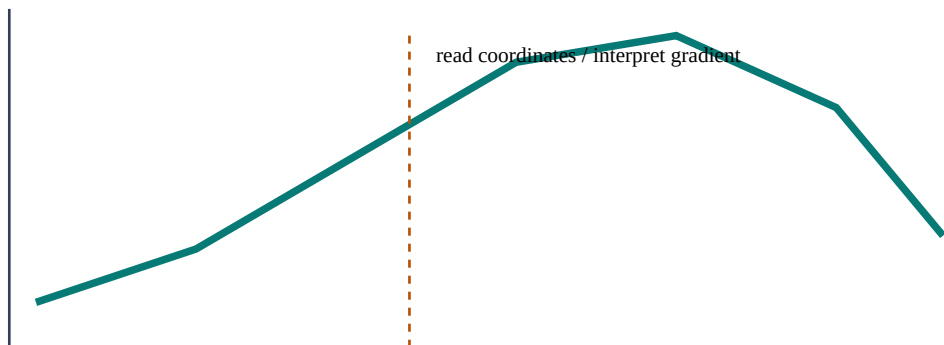
English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

实线表示 $\leq$ 或 $\geq$ ，虚线表示 $<$ 或 $>$ ；用测试点判断阴影侧。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
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4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

解 $3(2x+1) > 9$ 。

$6x+3 > 9$; $6x > 6$; 除以6后反向, $x < 1$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 24 课（续） | 不等式与区域 | Inequalities and regions

例题二：Cambridge 风格应用 | Cambridge-style application

Shade $y \leq 2x+1$ and $x > 0$.

Draw $y=2x+1$ as a solid line, $x=0$ as a dashed line; shade below the first and right of the second.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

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只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

不等式表示范围；乘除负数时符号反向。二维不等式用边界线和阴影区域表示。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。实线表示 $\leq$ 或 $\geq$ ，虚线表示 $<$ 或 $>$ ；用测试点判断阴影侧。

Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：解 $3(2x+1)>9$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $6x+3>9$ ； $6x>6$ ；除以6后反向， $x<1$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Shade $y \leq 2x+1$ and $x > 0$.

独立完成后对照：Draw $y=2x+1$ as a solid line, $x=0$ as a dashed line; shade below the first and right of the second.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 25 课 | 函数、反函数与复合函数 | Functions

Lesson 25 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

函数把每个允许输入映射到唯一输出；反函数撤销原函数，复合函数按从右到左作用。

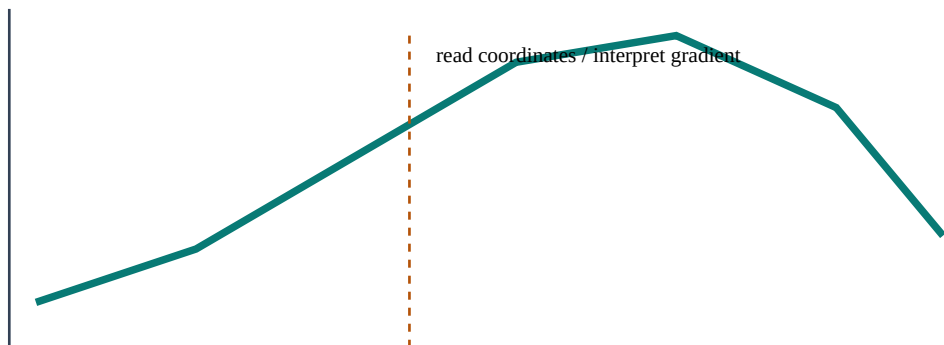
English core explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

为什么方法成立 | Reasoning behind the method

求反函数：写 $y=f(x)$ ，交换 x,y ，再解 y ；同时考虑定义域。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$f(x)=3x5$ ，求 $f(x)$ 和 $f(10)$ 。

$y=3x5$ ；交换得 $x=3y5$ ； $y=(x+5)/3$ 。因此 $f(10)=5$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 25 课（续） | 函数、反函数与复合函数 | Functions

例题二：Cambridge 风格应用 | Cambridge-style application

If $f(x)=x$ and $g(x)=x+1$, find $fg(3)$.

$fg(3)=f(g(3))=f(4)=16$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Detailed English explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

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第三个迁移任务 | Transfer task

If $f(x)=x$ and $g(x)=x+1$, find $fg(3)$.

独立完成后对照： $fg(3)=f(g(3))=f(4)=16$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 26 课 | 比例、变化率与增长衰减 | Variation and growth

Lesson 26 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

直接变化 $y \propto x$, 反比例 $y \propto 1/x$; 复合变化必须先写含常数 k 的方程。

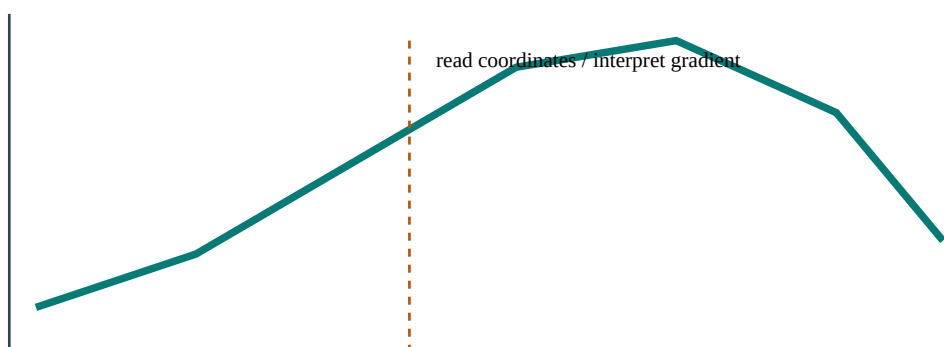
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

用一组数据求 k , 再代新数据; 增长衰减用反复乘同一倍率。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一：从题意到完整解答 | Worked example 1

y 与 x 成正比, $x=3$ 时 $y=18$, 求 $x=5$ 时 y 。

$y=kx$; $18=9k$, $k=2$; $y=2 \times 25=50$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 26 课（续） | 比例、变化率与增长衰减 | Variation and growth

例题二：Cambridge 风格应用 | Cambridge-style application

A value depreciates 12% yearly from 25 000. Find it after 4 years.

$25\,000(0.88) \approx 14\,993$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

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第三个迁移任务 | Transfer task

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独立完成后对照： $25\,000(0.88) \approx 14\,993$.

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第 27 课 | 圆定理 | Circle theorems

Lesson 27 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

圆周角、圆心角、切线与弦之间有固定关系；答题必须说明所用定理。

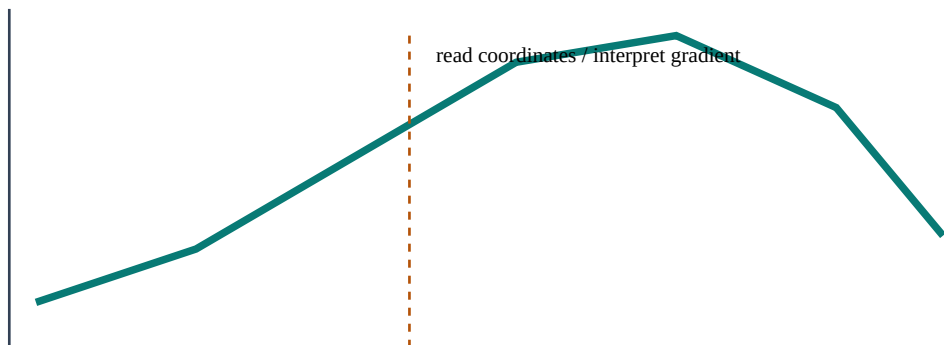
English core explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

为什么方法成立 | Reasoning behind the method

同弧圆心角是圆周角两倍；半圆圆周角 90° ；同弦同弓形角相等；切线垂直半径。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一：从题意到完整解答 | Worked example 1

弧AB的圆心角 $\angle AOB = 124^\circ$ ，求同弧圆周角 $\angle ACB$ 。

圆周角 $= 124^\circ \div 2 = 62^\circ$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 27 课（续） | 圆定理 | Circle theorems

例题二：Cambridge 风格应用 | Cambridge-style application

A tangent touches a circle at T and OT is a radius. Find angle between OT and tangent.

90° , because a tangent is perpendicular to the radius at the point of contact.

评分点如何产生 | Where marks come from

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：弧AB的圆心角 $\angle AOB = 124^\circ$ ，求同弧圆周角ACB。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：圆周角 $= 124^\circ \div 2 = 62^\circ$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” $\rightarrow$ “Using ..., we obtain ...” $\rightarrow$ “Substituting and simplifying gives ...” $\rightarrow$ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

A tangent touches a circle at T and OT is a radius. Find angle between OT and tangent.

独立完成后对照： 90° ，because a tangent is perpendicular to the radius at the point of contact.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 28 课 | 正弦、余弦定理与三维问题 | Advanced trigonometry

Lesson 28 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

非直角三角形用 sine rule、cosine rule 和面积公式 $ab \sin C$; 三维题先找到含目标的平面三角形。

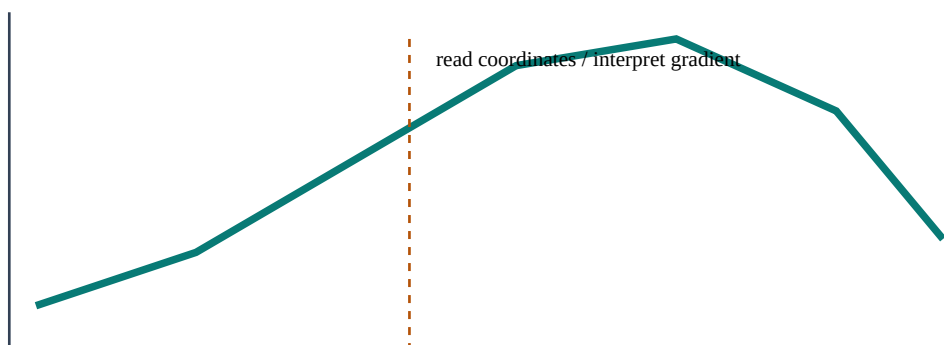
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

已知边角对应关系时用正弦定理; 三边或夹角用余弦定理。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

两边7、10夹角 48° ，求第三边。

$c = \sqrt{7^2 + 10^2 - 2 \times 7 \times 10 \cos 48^\circ}$, $c \approx 7.45$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 28 课（续） | 正弦、余弦定理与三维问题 | Advanced trigonometry

例题二：Cambridge 风格应用 | Cambridge-style application

Find area when $a=8$, $b=11$ and included angle is 35° .

Area = $\frac{1}{2} \times 8 \times 11 \times \sin 35^\circ \approx 25.2$ square units.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

非直角三角形用 sine rule、cosine rule 和面积公式 $ab \sin C$; 三维题先找到含目标的平面三角形。考试不会总以熟悉的标题出现: 它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”, 再决定使用哪条规则。已知边角对应关系时用正弦定理; 三边或夹角用余弦定理。

Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母, 应先建立一般关系再代值; 若改变范围、方向、图形性质或随机条件, 则必须重新检查原方法的适用条件。把答案遮住, 尝试只凭题干写出第一行模型; 第一行正确, 后续计算才有方向。

第三道示范: 审题、计算、复核 | Worked reasoning example

重新审视本课基础题: 两边7、10夹角 48° , 求第三边。

先把文字条件转换成数学关系, 再依据本课方法推进。完整结果是: $c = \sqrt{7^2 + 10^2 - 2 \times 7 \times 10 \cos 48^\circ}$, $c \approx 7.45$ 。

复核时从结论逆向代回, 并检查量级、正负、单位和定义域。若不一致, 优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项: 漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果, 再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

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第三个迁移任务 | Transfer task

Find area when $a=8$, $b=11$ and included angle is 35° .

独立完成后对照: $\text{Area} = \frac{1}{2} \times 8 \times 11 \times \sin 35^\circ \approx 25.2$ square units.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 29 课 | 矩阵与变换 | Matrices

Lesson 29 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

2×2 矩阵可表示平面线性变换; 列向量输入左乘矩阵得到像。

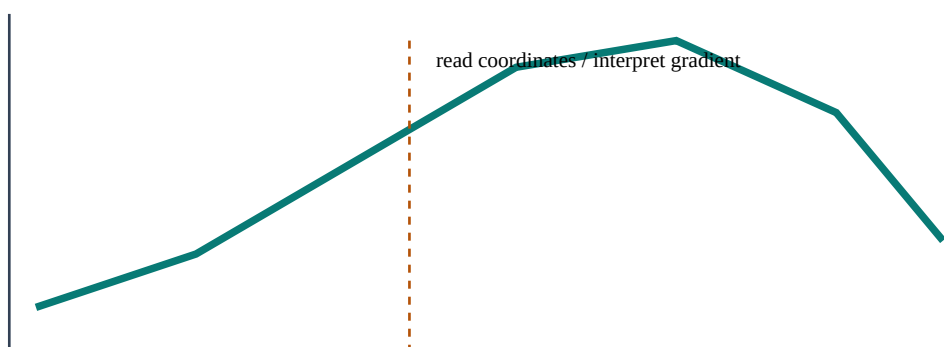
English core explanation. Direction and order matter. A transformation must be described with every defining parameter; vector subtraction gives directed displacement; matrix multiplication acts in a fixed order; complex modulus and argument encode size and direction.

为什么方法成立 | Reasoning behind the method

用基向量 $(1,0), (0,1)$ 的像构造矩阵两列; 矩阵乘法注意行乘列。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一: 从题意到完整解答 | Worked example 1

矩阵 $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ 作用于 $(3,2)$ 。

得到 $(2,3)$, 表示绕原点逆时针旋转 90° 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 29 课（续） | 矩阵与变换 | Matrices

例题二：Cambridge 风格应用 | Cambridge-style application

Find determinant of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

$$\det = 4 \times 3 - 1 \times 2 = 10.$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

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变式推演 | Variation and transfer

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

Find determinant of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

独立完成后对照： $\det = 4 \times 3 - 1 \times 2 = 10$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 30 课 | 概率树、条件概率 | Tree diagrams and conditional probability

Lesson 30 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

条件概率改变样本空间；不放回抽取会改变第二层分支概率。

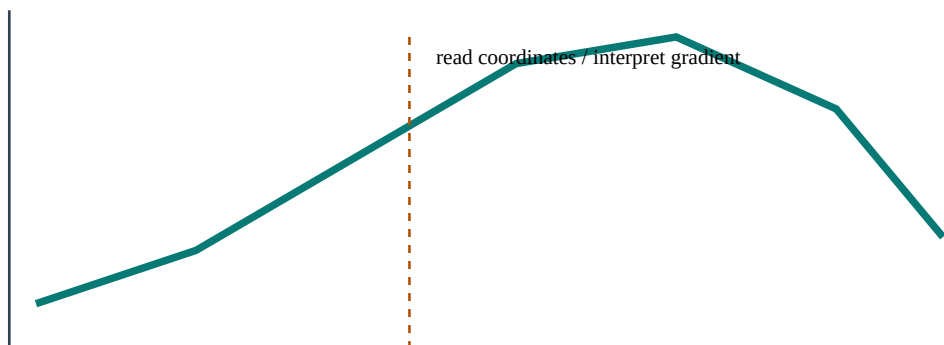
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

沿路径相乘，互斥路径相加； $P(A|B)=P(A \cap B)/P(B)$ 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一：从题意到完整解答 | Worked example 1

盒中4白2黑，不放回抽2个，求恰好1白。

$$4/6 \times 2/5 + 2/6 \times 4/5 = 8/15。$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 30 课（续） | 概率树、条件概率 | Tree diagrams and conditional probability

例题二：Cambridge 风格应用 | Cambridge-style application

Given $P(A \cap B) = 0.18$ and $P(B) = 0.3$, find $P(A|B)$.

$$0.18 \div 0.3 = 0.6.$$

评分点如何产生 | Where marks come from

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易错辨析 | Common errors and repairs

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Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

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英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Given $P(A \cap B)=0.18$ and $P(B)=0.3$, find $P(A|B)$.

独立完成后对照： $0.18 \div 0.3=0.6$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 31 课 | 直方图与统计比较 | Histograms and comparison

Lesson 31 of 31 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

组距不等时柱高是frequency density, 不是frequency; 面积代表频数。

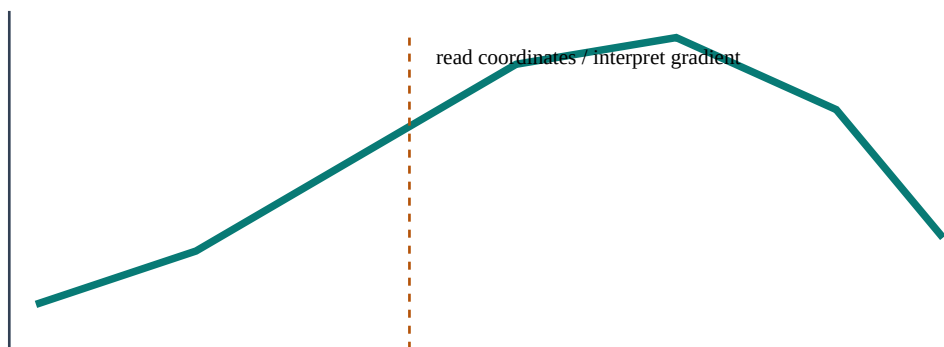
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

frequency density=frequency/class width. 比较分布要同时比较中心和离散。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
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5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

组 $20 \leq x < 30$ 频数24, 求频数密度。

组距10, 密度=24/10=2.4。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 31 课（续） | 直方图与统计比较 | Histograms and comparison

例题二：Cambridge 风格应用 | Cambridge-style application

Class A has lower median and smaller IQR than B. Interpret.

A has a lower typical value and is more consistent because its middle 50% is less spread out.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

组距不等时柱高是frequency density, 不是frequency; 面积代表频数。考试不会总以熟悉的标题出现: 它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”, 再决定使用哪条规则。 $\text{frequency density} = \text{frequency} / \text{class width}$ 。比较分布要同时比较中心和离散。

Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母, 应先建立一般关系再代值; 若改变范围、方向、图形性质或随机条件, 则必须重新检查原方法的适用条件。把答案遮住, 尝试只凭题干写出第一行模型; 第一行正确, 后续计算才有方向。

第三道示范: 审题、计算、复核 | Worked reasoning example

重新审视本课基础题: 组 $20 \leq x < 30$ 频数24, 求频数密度。

先把文字条件转换成数学关系, 再依据本课方法推进。完整结果是: 组距10, 密度 $=24/10=2.4$ 。复核时从结论逆向代回, 并检查量级、正负、单位和定义域。若不一致, 优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项: 漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果, 再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” $\rightarrow$ “Using ..., we obtain ...” $\rightarrow$ “Substituting and simplifying gives ...” $\rightarrow$ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Class A has lower median and smaller IQR than B. Interpret.

独立完成后对照: A has a lower typical value and is more consistent because its middle 50% is less spread out.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?